सी एस आई आर - राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
AN ELECTRONIC BALANCE

प्रमाण पत्र संख्या / **Certificate No.**

25070438/D1.01/C-086

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
07.10.2025	07.10.2028	1	2

1. Calibrated for : National Council for Cement & Building Materials
34 km Stone, Delhi Mathura Road
Ballabhgarh - 121004, Harayana, India

Customer's Ref. No. : Letter Dated : 01.07.2025

2. Description and identification of instrument : An electronic balance with specifications :

Make	:	Sartorius
Model	:	MCM6.7
Serial No.	:	46601101
Weighing capacity	:	6.1 g
Readability (d)	:	0.0001 mg

3. Environmental conditions :

Temperature	:	(23 ± 1) °C
Relative humidity	:	(50 ± 10) %

4. Standard(s) used with associated Measurement uncertainty : NPL working standards (Sub-Div.#1.01/M/224(2)/00 & Sub-Div.#1.01/M/60027/55 of mass with uncertainty better than one-third of the reported uncertainty of measurement.

5. Metrological Traceability of Standard(s) used : The working standard(s) used for calibration is(are) traceable to the National Standard which realize the physical units of mass according to the International System of units (SI).

6. Principle/Methodology of Calibration & Calibration Procedure number : CSIR-NPL Calibration Procedure No. Sub-Div.#1.01/Doc.3/CP#WI/M-04. The calibration procedure is based on OIML R 76-1 : 2006.

अंशांकनकर्ता: Calibrated by:

जाँचकर्ता: Checked by:

प्रभारी वैज्ञानिक: Scientist-in-charge:

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(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
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CALIBRATION CERTIFICATE

AN ELECTRONIC BALANCE

प्रमाण पत्र संख्या / Certificate No.

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07.10.2025	07.10.2028	2	2

7. Results :

i. Weighing performance test

The balance was tested for its accuracy, both for progressively increasing and decreasing loads. The indication errors at diffrent loads were found to be as follows:

Denomination	Error (mg)	MPE (mg)
2 mg	0.000 7	± 0.500 0
5 mg	0.000 4	± 0.500 0
10 mg	0.000 5	± 0.500 0
20 mg	0.000 6	± 0.500 0
50 mg	0.000 5	± 0.500 0
100 mg	0.000 7	± 0.500 0

Denomination	Error (mg)	MPE (mg)
200 mg	0.001 0	± 0.500 0
500 mg	0.000 7	± 0.500 0
1 g	0.000 9	± 0.500 0
2 g	0.000 4	± 0.500 0
5 g	0.000 3	± 0.500 0

ii. Repeatability test

The balance was tested for repeatability of its indication at 1 g, 2 g & 5 g loads. The differences between the maximum and minimum indications for the above loads were found to be 0.001 5 mg, 0.001 5 & 0.001 8 mg respectively against the maximum permissible errors of ± 0.500 0 mg, ± 0.500 0 mg & ± 0.500 0 mg respectively. The standard deviations were found to be 0.000 4 mg, 0.000 5 mg & ± 0.000 5 mg respectively.

iii. Uncertainty of measurement :

Denomination	Uncertainty (mg)
5 g to 1 g	± 0.001 0
500 mg to 100 mg	± 0.000 7
50 mg to 1 mg	± 0.000 5

The reported uncertainty is at coverage factor k = 2 which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. Date(s) of Calibration : 01.08.2025

9. Remarks :
1. Balance is calibrated at site (Mass Laboratory, NCCBM, Ballabhgarh).
2. Environmental conditions were provided by the customer.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)